$$\text{A} = \begin{matrix} \\ \text{Thilagan} \\ \text{Kausigan} \end{matrix} \begin{matrix} \text{April sale in ₹} \\ \begin{matrix} \text{rice} & \text{wheat} & \text{ragi} \end{matrix} \\ \begin{pmatrix} 500 & 1000 & 1500 \\ 2500 & 1500 & 500 \end{pmatrix} \end{matrix}$$

and the May month sale (in ₹) is exactly twice as that of the April month sale for each variety.

(i) What is the average sales of the months April and May.

(ii) If the sales continues to increase in the same way in the successive months, what will be sales in the month of August?

18. If $\cos\theta \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} + \sin\theta \begin{pmatrix} x & -\cos\theta \\ \cos\theta & x \end{pmatrix} = I_2$, find x.

19. Given $A = \begin{pmatrix} p & 0 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -q \\ 1 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$ and if $BA = C^2$, find p and q.

20. $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 3 \\ 8 & 5 \end{pmatrix}$, $C = \begin{pmatrix} 3 & 6 \\ 1 & 1 \end{pmatrix}$ find the matrix D, such that $CD - AB = 0$

Points to Remember

- A system of linear equations in three variables will be according to one of the following cases.

  (i) Unique solution (ii) Infinitely many solutions (iii) No solution
- The least common multiple of two or more algebraic expressions is the expression of lowest degree (or power) such that the expressions exactly divides it.
- A polynomial of degree two in variable x is called a quadratic polynomial in x. Every quadratic polynomial can have atmost two zeroes. Also the zeroes of a quadratic polynomial intersects the x-axis.
- The roots of the quadratic equation $ax^2 + bx + c = 0$, $(a \neq 0)$ are given by $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- For a quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$

  Sum of the roots $\alpha + \beta = \frac{-b}{a} = \frac{-\text{Co-efficient of } x}{\text{Co-efficient of } x^2}$

  Product of the roots $\alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Co-efficient of } x^2}$
- If the roots of a quadratic equation are α and β, then the equation is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

3 Algebra.indd 158 16-12-2020 01:08:09 PM